

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II (NEP) Examination 2024-25

DEPARTMENT OF PHYSICS**Paper Code: PHYMJ21 / PHYMN21 — Thermal Physics**

Time: 2:30 Hours

Maximum Marks: 70

Paper Code Ref: CECNF/N/2024-25/4

*Instruction: Write your Roll No. at the top immediately on the receipt of this question paper.**Note: Attempt five questions including Question 1 which is compulsory. Question 1 carries 20 marks, and Questions 2–7 carry 12.5 marks each.***Question 1** Answer all questions briefly:

[10 × 2 = 20]

- (a) State the zeroth law of thermodynamics.
- (b) Explain Zartman and Ko's experiment for verification of Maxwell's law of distribution of speeds.
- (c) Calculate the mean free path for molecules of hydrogen gas. The radius of the gas molecule is 1.37×10^{-10} m and the density of gas is 3×10^{25} molecules/m³.
- (d) Write photon gas pressure inside a cavity in terms of photon's energy density.
- (e) Draw T - S diagram of a reversible Carnot cycle.
- (f) Calculate the change in entropy when 1 kg of water at 100°C is converted into steam at the same temperature. Latent heat of steam is 540 cal/g.
- (g) Write inversion temperature in terms of real gas constants.
- (h) Write the relation between Gibbs and Helmholtz free energy.
- (i) Explain the triple point of water using a suitable diagram.
- (j) Explain Joule-Thomson effect.

Question 2

- (a) Explain the Carnot cycle. Show that no irreversible heat engine operating between two reservoirs (source and sink having absolute temperatures T_1 and T_2 respectively) can be more efficient than a reversible engine operating between the same reservoirs. [8.5]
- (b) Calculate the work done in kJ during an isothermal reversible process when 5 moles of ideal gas are expanded so that its volume is doubled at a temperature of 400 K. [4]

Question 3

- (a) Show that the change in entropy is zero for a reversible process, but it increases for an irreversible process. [5.5]
- (b) Write the relation between heat capacities at constant pressure (C_p) and at constant volume (C_v) as a function of isothermal compressibility and volume expansivity. Calculate the difference between them for Helium gas at temperature 6 K and pressure 19.7 atm. (Given: specific volume = $2.64 \times 10^{-2} \text{ m}^3/\text{kmol}$, isothermal compressibility = $9.42 \times 10^{-8} \text{ m}^2/\text{N}$, volume expansivity = $5.35 \times 10^{-2} \text{ K}^{-1}$). [7]
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Question 4

- (a) Derive Clausius-Clapeyron equation using Carnot's cycle. [7.5]
- (b) Prove the following thermodynamic relation:

$$TdS = C_p dT - T \left(\frac{\partial V}{\partial T} \right)_P dP$$

[5]

Question 5

- (a) Derive the Rayleigh-Jeans law. Also explain the UV catastrophe. [7.5]
- (b) The Sun's radius is $6.96 \times 10^8 \text{ m}$ and its surface temperature is 5700 K. Estimate the total power radiated by the Sun. ($\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$). [5]
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Question 6

- (a) Explain transport phenomena in gases. Derive the expression for viscosity of gases. [7.5]
- (b) Write critical gas constants in terms of real gas constants. Find the gas constant b if critical volume V_c is 69.7 cc/mole. [5]
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Question 7

- (a) Derive the following Maxwell's thermodynamic relation:

$$\left(\frac{\partial T}{\partial P} \right)_S = \left(\frac{\partial V}{\partial S} \right)_P$$

[7.5]

- (b) Explain the principle of regenerative cooling with a schematic diagram. [5]

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